

CHAPTER 3 km reanalysis — GRIB2 archive contents

90 variables, 246 GRIB messages per hourly file. Taken from the converter's own registry (wrf_era5_comparison.WRF_TO_ECMWF_PARAMID), which is what the pipeline and its sanity check use, so this document describes what is actually written. Document updated on 2026-09-18, when the WPS static file geo_em.d02 was recovered from LRZ after having been recorded as lost. It carries the FRACTIONAL land cover, soil texture and lake data that the hourly output had reduced to a single dominant category per cell, and it added fifteen fields: cvl, cvh, tvl, tvh, slt, cl, dl and the eight ERA5 soil layers. Nothing was removed. Every field here is one we can defend row by row. Where the WRF quantity is not exactly the ERA5 one, it is said so below rather than hidden; where it could not be made to correspond at all, the field was dropped instead of published under a name that would mislead (see §6).

1. What the files are

Model	WRF 4.1.1, domain d02, 49 vertical levels, model top 50 hPa
Horizontal grid	Mercator, 1641 × 1353 points, 3 km spacing (2 220 273 points/field)
Domain corners	23.472 N, 19.972 W → 60.001 N, 42.256 E; standard parallel 44.671 N
Earth shape	Sphere of radius 6 370 000 m (shapeOfTheEarth=1) — WRF's own sphere, declared explicitly
GRIB edition	GRIB2 , grid definition template 3.10 (Mercator), originating centre ecmf
Packing	grid_ccsds, 24 bits per value ; masked fields carry a proper GRIB bitmap
Parameter identity	ECMWF paramId / shortName throughout, so the archive is directly comparable with ERA5
File	one file per hour, ~747 MB (~17.9 GB/day, ~6.6 TB/year) — measured, not estimated
Naming	ailam-an-cima-3km- <code>{year}-{year}-1h-v1-<code>{YYYYMMDD}</code>{HH}</code> .grib
Period	2019, 2024 and 2025-01-01 → 2025-06-30, hourly (2020–2023 to follow)

2. Vertical levels

13 pressure levels, the same for every 3-D variable:

1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa

They are the ERA5 set truncated at the model top: the 8 ERA5 levels above 50 hPa are above WRF's P_TOP and cannot be produced. Single-level fields sit on surface, heightAboveGround (2, 10, 100 and 200 m), meanSea, entireAtmosphere (column integrals) or, for the radiation budget at the top of the atmosphere, nominalTop.

3. Time convention — read this before indexing the archive

- **Instantaneous fields** use product definition template **0**: dataDate/dataTime are the valid time.
- **Accumulated fields** (tp, tirf, sf, sro, ssro, ro, and the whole radiation budget) use template

8, as their ECMWF definitions require. They are accumulated ****from 00 UTC of the same day****: reference time 00 UTC, step 0–H. They are exactly zero in the 00 UTC file.

- **10fg** is the one exception: it is a **maximum over the preceding hour** (reference time H–1, step 0–1), and it is the maximum of the *resolved* 10 m wind, not a gust parameterisation.
- A file therefore mixes reference times on purpose. ****Index on validity (validityDate/validityTime), never on dataTime.****
- Accumulated messages carry generatingProcessIdentifier = 128 as the marker of the 00 UTC reference.

4. Declared approximations

Three fields are not exactly their ERA5 namesake, and are published with that said:

field	what ERA5 means	what we write
slor	slope of the sub-grid orography, a parameter of the form-drag scheme	the slope of the resolved 3 km terrain, $ \text{grad } z $
10fg	gust from a parameterisation (turbulent + convective)	the hourly maximum of the resolved 10 m wind
IEWS, INSS	the model's own stress components	the magnitude ρu^2 , projected on the 10 m wind direction

Three more deserve a note rather than a warning:

- **skt** is not a model output here (TSK was not written): it is inverted from the upward longwave, $LWUPB = \epsilon \sigma T^4 + (1-\epsilon) LWDN$, with the emissivity of the dominant land-use category. It is exact where the answer is known — over open water the model's skin temperature is the SST, and the inversion recovers it to **0.0015 K**. Which emissivity the model used was settled by measurement rather than assumed: the relation is linear in $\sigma T^4 - LWDN$, so its slope is the emissivity, and fitted cell by cell over a full diurnal cycle it comes out a per-category constant that July and March agree on to 0.0005. Agreement with the 0 cm soil level on snow-free land, by surface type:

surface	night	day
forests, savanna, grassland, urban (7 classes)	0.03–0.11 K RMSE	0.09–0.22 K
mixed forest, cropland	0.12–0.19 K	0.21–0.30 K
open shrubland	0.72 K (bias –0.68)	1.22 K (bias –1.15)
barren / sparsely vegetated	0.75 K (bias –0.74)	1.54 K (bias –1.52)

Over the two arid classes **skt** therefore runs about 1–1.5 K colder than the soil reading. The emissivity fit there wants a value near 0.85, which would close the gap, but that is below every emissivity in every WRF table, so it cannot be what the model used and was not adopted. Over snow-covered points the scheme's own emissivity cannot be reconstructed at all.

- **sst** is constant within each 24 h run (SST_UPDATE=0), re-initialised daily.
- **dL** is the lake depth database the run was built with, not a model state: lake physics was off (sf_lake_physics=0), so nothing in the simulation responds to it. It is published as the boundary dataset it is, and masked off-lake — the raw field is the 10 m WPS default fill on 99.56 % of the domain, and 79 464 of the 88 785 lake cells carry that same default (ERA5's own **dL** does the same where the database is silent).

5. Masked fields

Written with a GRIB bitmap rather than fake zeros, so missing means missing:

field	defined where
sst, ci	sea points only
tsn	cells whose snow cover exceeds 0.9 (≈ 29 mm water equivalent); out of season the field is legitimately empty
fal	daytime only (incoming shortwave above 50 W/m ²)
swvl1-4, stl1-4	land points only, where the land-surface scheme integrates a soil column. This mask moves: the sea-ice reclassification turns about 2 500 sea points into land between seasons, so it follows the hourly l_{sm} , not a fixed outline
slt	land points of the static file (1 304 109 cells)
tv1	everywhere except the cells where a shrubland dominates the low vegetation (124 672 cells, 15.8 % of those with $cv1 > 0$). MODIS carries no shrub phenology while ECMWF's table splits evergreen from deciduous shrubs, and rather than invent one the field is left missing. $cv1$ still gives the cover there
dl	lake cells only (88 785)

6. What is not in the archive

No TKE and no boundary-layer height (the PBL scheme is YSU, non-local); no ocean waves, currents, sea surface height or sea-ice thickness (no ocean or wave coupling); no convective precipitation (CU_PHYSICS=0, so it is identically zero and would be a dead field). Deliberately dropped rather than approximated:

- **leaf area index** (lai_{lv} , lai_{hv}): the model carries a single LAI per cell, while ERA5 wants one per vegetation class, defined per unit of vegetated area. Any split between the two would be our assumption, so neither field is written. The model's own LAI can be delivered under its own name if it is wanted;
- **sub-grid orography angle and anisotropy** ($anor$, $isor$): the static file carries the Kim-Arakawa asymmetry and effective-length parameters, which are not ECMWF's, and the difference does not fit in one line.

Four entries that stood here until 2026-09-17 have moved into the archive, because `geo_em.d02` came back and because the soil objection had an answer:

- **swvl1-4 / stl1-4** are now the true ERA5 layer averages. The land-surface scheme is a *level* scheme — point values at 0, 5, 20, 40, 160 and 300 cm with a linear profile between them — and the whole ERA5 column lies inside those nodes (289 cm is above the deepest, 0 cm is the first). So each layer average is the exact integral of the model's own profile and **nothing is extrapolated**. The objection was against writing the point values, not against integrating them;
- **cv1 / cvh** are sums of the fractional cover over the MODIS-IGBP classes that are low and high vegetation — the same quantity ERA5 means. This is what the dominant category could not express: it holds only 90.1 % of a land cell on average, and 17.9 % of land cells carry both high and low vegetation above 5 %;

- **tv1 / tvh** are the dominant low and high class translated to ECMWF code table 4.234 by the table in §8, where each row is a semantic identity. The one place MODIS cannot answer the question — evergreen versus deciduous shrubs — is left missing rather than guessed;
- **slt** is computed with the FAO clay/sand thresholds that *define* ECMWF’s seven soil types, applied to the static file’s own clay and sand fractions. No category crosswalk is involved, so the quantity is ERA5’s by construction. One caveat for a strict WMO reader: eccodes resolves **slt** to discipline 2/3/0, which WMO table 4.213 reads as a different 11-value list. This archive is ECMWF semantics throughout, and so is this message.

cl and **dl** came with them. **cl** is not redundant with the land-sea mask: 19 309 cells that the mask calls land carry a sub-grid lake, and the model itself discarded the class at run-time. The full reasoning, and the routes by which some of it could still be obtained, is in `MISSING_VARIABLES.pdf`.

7. The variables

Descriptions and units are the ECMWF ones. “Time” says how the field is sampled. There are 90 rows and 89 distinct shortNames, because **z** is written twice: as geopotential on the pressure levels and as the surface geopotential (orography). variables=90 messages=246

Pressure levels (13) — 13 variables, 169 messages

shortName	paramId	Description	Units	Level	Time
cc	248	Fraction of cloud cover	(0-1)	13 levels	instantaneous
ciwc	247	Specific cloud ice water content	kg/kg	13 levels	instantaneous
clwc	246	Specific cloud liquid water content	kg/kg	13 levels	instantaneous
crwc	75	Specific rain water content	kg/kg	13 levels	instantaneous
cswc	76	Specific snow water content	kg/kg	13 levels	instantaneous
q	133	Specific humidity	kg/kg	13 levels	instantaneous
r	157	Relative humidity	%	13 levels	instantaneous
t	130	Temperature	K	13 levels	instantaneous
u	131	U component of wind	m/s	13 levels	instantaneous
v	132	V component of wind	m/s	13 levels	instantaneous
w	135	Vertical velocity (pressure)	Pa/s	13 levels	instantaneous
wz	260238	Geometric vertical velocity	m/s	13 levels	instantaneous
z	129	Geopotential	m ² /s ²	13 levels	instantaneous

Height above ground — 11 variables, 11 messages

shortName	paramId	Description	Units	Level	Time
100u	228246	100m U component of wind	m/s	100 m	instantaneous
100v	228247	100m V component of wind	m/s	100 m	instantaneous
10fg	49	10m wind gust (hourly max of resolved wind)	m/s	10 m	max over the previous hour
10u	165	10m U component of wind	m/s	10 m	instantaneous
10v	166	10m V component of wind	m/s	10 m	instantaneous
200u	228239	200m U component of wind	m/s	200 m	instantaneous
200v	228240	200m V component of wind	m/s	200 m	instantaneous
2d	168	2m dewpoint temperature	K	2 m	instantaneous
2r	260242	2m relative humidity	%	2 m	instantaneous
2t	167	2m temperature	K	2 m	instantaneous
vwsh	260068	Vertical speed shear (bulk 10-100 m)	s ⁻¹	100 m	instantaneous

Mean sea level — 1 variables, 1 messages

shortName	paramId	Description	Units	Level	Time
msl	151	Mean sea level pressure	Pa	mean sea level	instantaneous

Total column — 10 variables, 10 messages

shortName	paramId	Description	Units	Level	Time
hcc	188	High cloud cover	(0-1)	column	instantaneous
lcc	186	Low cloud cover	(0-1)	column	instantaneous
mcc	187	Medium cloud cover	(0-1)	column	instantaneous
tcc	164	Total cloud cover	(0-1)	column	instantaneous

shortName	paramId	Description	Units	Level	Time
tciw	79	Total column cloud ice water	kg/m ²	column	instantaneous
tclw	78	Total column cloud liquid water	kg/m ²	column	instantaneous
tcrw	228089	Total column rain water	kg/m ²	column	instantaneous
tcsw	228090	Total column snow water	kg/m ²	column	instantaneous
tcw	136	Total column water	kg/m ²	column	instantaneous
tcwv	137	Total column water vapour	kg/m ²	column	instantaneous

Surface / single level — 47 variables, 47 messages

shortName	paramId	Description	Units	Level	Time
al	174	Albedo (climato-logical, snow-free background)	(0-1)	surface	instantaneous
ci	31	Sea ice area fraction	(0-1)	surface	instantaneous
cl	26	Lake cover	(0-1)	surface	instantaneous, static
cvh	28	High vegetation cover	(0-1)	surface	instantaneous, static
cvl	27	Low vegetation cover	(0-1)	surface	instantaneous, static
fal	243	Forecast albedo (upward/downward SW at surface)	(0-1)	surface	instantaneous
fsr	244	Forecast surface roughness	m	surface	instantaneous
iews	229	Instantaneous eastward turbulent surface stress	N/m ²	surface	instantaneous
inss	230	Instantaneous northward turbulent surface stress	N/m ²	surface	instantaneous
lsm	172	Land-sea mask	(0-1)	surface	instantaneous
ro	205	Runoff (surface + sub-surface)	m	surface	accumulated since 00 UTC
rsn	33	Snow density	kg/m ³	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
sd	141	Snow depth (water equivalent)	m	surface	instantaneous
sdor	160	Standard deviation of orography	m	surface	instantaneous
sf	144	Snowfall (water equivalent)	m	surface	accumulated since 00 UTC
skt	235	Skin temperature (inverted from upward longwave)	K	surface	instantaneous
slor	163	Slope of orography (resolved, 3 km)	Numeric	surface	instantaneous
slt	43	Soil type (ECMWF 7 classes)	~	surface	instantaneous, static
snowc	260038	Snow cover	%	surface	instantaneous
sp	134	Surface pressure	Pa	surface	instantaneous
src	198	Skin reservoir content	m	surface	instantaneous
sro	8	Surface runoff	m	surface	accumulated since 00 UTC
ssr	176	Surface net solar radiation	J/m ²	surface	accumulated since 00 UTC
ssrc	210	Surface net solar radiation, clear sky	J/m ²	surface	accumulated since 00 UTC
ssrd	169	Surface solar radiation downwards	J/m ²	surface	accumulated since 00 UTC
ssrdc	228129	Surface solar radiation downwards, clear sky	J/m ²	surface	accumulated since 00 UTC
ssro	9	Sub-surface runoff	m	surface	accumulated since 00 UTC
sst	34	Sea surface temperature	K	surface	instantaneous
stl1	139	Soil temperature level 1 (0-7 cm)	K	surface	instantaneous
stl2	170	Soil temperature level 2 (7-28 cm)	K	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
stl3	183	Soil temperature level 3 (28-100 cm)	K	surface	instantaneous
stl4	236	Soil temperature level 4 (100-289 cm)	K	surface	instantaneous
str	177	Surface net thermal radiation	J/m ²	surface	accumulated since 00 UTC
strc	211	Surface net thermal radiation, clear sky	J/m ²	surface	accumulated since 00 UTC
strd	175	Surface thermal radiation downwards	J/m ²	surface	accumulated since 00 UTC
strdc	228130	Surface thermal radiation downwards, clear sky	J/m ²	surface	accumulated since 00 UTC
swvl1	39	Volumetric soil water layer 1 (0-7 cm)	m ³ /m ³	surface	instantaneous
swvl2	40	Volumetric soil water layer 2 (7-28 cm)	m ³ /m ³	surface	instantaneous
swvl3	41	Volumetric soil water layer 3 (28-100 cm)	m ³ /m ³	surface	instantaneous
swvl4	42	Volumetric soil water layer 4 (100-289 cm)	m ³ /m ³	surface	instantaneous
tirf	235015	Time integral of rain flux (rain only)	kg/m ²	surface	accumulated since 00 UTC
tp	228	Total precipitation	m	surface	accumulated since 00 UTC
tsn	238	Temperature of snow layer	K	surface	instantaneous
tvh	30	Type of high vegetation (ECMWF code table 4.234)	~	surface	instantaneous, static
tvL	29	Type of low vegetation (ECMWF code table 4.234)	~	surface	instantaneous, static
z	129	Geopotential (surface)	m ² /s ²	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
zust	228003	Friction velocity	m/s	surface	instantaneous

“static” in the Time column means the field is identical in every file of the archive: it comes from the run’s static input, not from the hourly state. `lsm` is deliberately NOT static — the model reclassifies sea points as land where sea ice appears, so it changes with the season.

Entire lake — 1 variable, 1 message

shortName	paramId	Description	Units	Level	Time
dl	228007	Lake total depth	m	entireLake	instantaneous, static

Top of atmosphere / most unstable parcel — 7 variables, 7 messages

shortName	paramId	Description	Units	Level	Time
mucape	228235	Most-unstable CAPE	J/kg	most unstable parcel	instantaneous
mucin	228236	Most-unstable CIN	J/kg	most unstable parcel	instantaneous
tisr	212	TOA incident solar radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
tsr	178	Top net solar radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
tsrc	208	Top net solar radiation, clear sky	J/m ²	top of atmosphere	accumulated since 00 UTC
ttr	179	Top net thermal radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
ttrc	209	Top net thermal radiation, clear sky	J/m ²	top of atmosphere	accumulated since 00 UTC

8. How the land-surface categories were translated

`tvL`, `tvh` and `sLt` are the only fields in the archive that carry a *code* rather than a number, so the tables that produced them are published here: without them the codes cannot be read, and the mapping cannot be checked. The source is the run’s own static file `geo_em.d02` (MODIS-IGBP 21 classes, WPS `MMINLU=MODIFIED_IGBP_MODIS_NOAH`). Its land use was verified against the hourly output: it agrees with the model’s dominant category on **100.00 %** of points once one expected difference is excluded — the model recodes lake to water at runtime, because lake physics was off. This is what makes `cL` recoverable at all.

`tvL` / `tvh`: MODIS-IGBP → ECMWF code table 4.234

High and low follow the IGBP class definitions of tree cover, not a canopy-height table.

MODIS-IGBP class	set	ECMWF 4.234
1 Evergreen Needleleaf Forest	high	3 Evergreen needleleaf trees

MODIS-IGBP class	set	ECMWF 4.234
2 Evergreen Broadleaf Forest	high	6 Evergreen broadleaf trees
3 Deciduous Needleleaf Forest	high	4 Deciduous needleleaf trees
4 Deciduous Broadleaf Forest	high	5 Deciduous broadleaf trees
5 Mixed Forests	high	18 Mixed forest/woodland
8 Woody Savannas	high	19 Interrupted forest (both are 30–60 % tree cover)
9 Savannas	low	7 Tall grass
10 Grasslands	low	2 Short grass
11 Permanent Wetlands	low	13 Bogs and marshes (absent from this domain)
12 Croplands	low	1 Crops, mixed farming
14 Cropland/Natural Mosaic	low	1 Crops, mixed farming (absent from this domain)
18 Wooded Tundra, 19 Mixed Tundra, 20 Barren Tundra	low	9 Tundra
6 Closed Shrublands, 7 Open Shrublands	low	not translated — tvl is missing there
13 Urban, 15 Snow and Ice, 16 Barren, 17 Water, 21 Lake	neither	— (not vegetation; they are the remainder of 1 – cvl – cvh)

Every translated row is a semantic identity: the ECMWF type names the same vegetation as the MODIS class. The two shrubland classes are the exception and are the reason tvl carries a mask. MODIS does not record whether a shrubland is evergreen or deciduous, while table 4.234 has separate codes for the two (16 and 17). Choosing one would have been our invention over 16.2 % of the vegetated cells, so the field is left missing there instead; cvl still reports the cover. The classes that would have been the other ambiguous rows, 11 and 14, do not occur in this domain at all.

slt: computed, not translated

No category crosswalk is involved. ECMWF's seven soil types are *defined* by clay and sand percentage thresholds, and the static file carries clay and sand fractions, so the definition is applied directly:

condition (percent)	ECMWF soil type
clay $\geq$ 60	5 very fine
$35 \leq$ clay < 60	4 fine
clay < 35 and sand $\leq$ 15	3 medium fine
($18 \leq$ clay < 35 and sand > 15) or (clay < 18 and $15 <$ sand $\leq$ 65)	2 medium
clay < 18 and sand > 65	1 coarse
dominant soil category is <i>organic material</i>	6 organic

Every land point classifies; type 7 (tropical organic) does not occur here. Distribution over the 1 304 109 land points: 1 069 456 medium, 187 378 coarse, 45 450 fine, 1 498 organic, 327 very fine.

swvl1-4 / stl1-4: integrated, not relabelled

The land-surface scheme is a *level* scheme: it carries point values at 0, 5, 20, 40, 160 and 300 cm, with a linear profile between them. ERA5 wants averages over 0–7, 7–28, 28–100 and 100–289 cm. Those four intervals lie entirely inside the model’s own nodes — 289 cm is above the deepest one and 0 cm is the first — so each ERA5 layer is the **exact integral** of the model’s profile over it, and nothing is extrapolated at either end. In practice this is a fixed 4×6 matrix of weights, applied to the six model levels. swvl uses the total soil moisture (liquid plus ice), which is what the ERA5 name means; the liquid-only field is not used. Both sets are masked over water.